Differential Forms: Pullback, Boundary Signs, and Stokes' Theorem

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The fundamental theorem of calculus relates an integral of a derivative to an expression on the boundary. Green's theorem, the divergence theorem and the classical Stokes theorem have the same structure, but their coordinates obscure it. Following Victor Guillemin's MIT 18.952 *Theory of Differential Forms* notes, we build the formula from pullback and exterior differentiation before stating it on a manifold.[1]

§1.1 A one-form eats a velocity vector

A covector at p is a linear function $T_pM \to \mathbb{R}$; a smooth one-form varies such covectors with p. In the plane write

$$\omega = P(x,y)\,dx + Q(x,y)\,dy.$$

For a curve $\gamma(t) = (x(t), y(t))$, the velocity is $\gamma'(t)$ and

$$\omega_{\gamma(t)}(\gamma'(t)) = P(x(t),y(t))x'(t) + Q(x(t),y(t))y'(t).$$

The pullback by γ packages this into a form on the parameter interval:

$$\gamma^*\omega = (P(\gamma(t))x'(t) + Q(\gamma(t))y'(t))\,dt.$$

Consequently $\int_\gamma \omega = \int_a^b \gamma^*\omega$. This is precisely the usual line-integral formula; it includes the chain rule and the orientation automatically.

For example, on the unit circle traversed counterclockwise,

$$\gamma(t) = (\cos t, \sin t), \qquad \gamma^*(x\,dy - y\,dx) = dt,$$

so $\int_\gamma (x\,dy - y\,dx) = 2\pi$. Reversing the parametrization changes the sign.

§1.2 An oriented area needs an alternating form

The wedge product is alternating:

$$dx \wedge dy = -dy \wedge dx, \qquad dx \wedge dx = 0.$$

For tangent vectors $u = (u_1, u_2)$, $v = (v_1, v_2)$, it evaluates as

$$(dx \wedge dy)(u, v) = u_1v_2 - u_2v_1 = \det[u\ v].$$

The determinant detects both oriented area and the order of the two vectors. Hence $F(x,y)\,dx \wedge dy$ is an appropriate integrand for an oriented surface, unlike an arbitrary scalar density stripped of its orientation.

A metric measures length by $ds = \|\gamma'(t)\|\,dt$. This is not the pullback of an ordinary one-form depending linearly on the velocity, because the norm satisfies $\|-v\| = \|v\|$. Reversing a curve changes a one-form integral's sign but leaves its length unchanged.

[1] https://math.mit.edu/classes/18.952/2015SP/materials.html

§1.3 Exterior differentiation contains the planar curl

For a function f, define $df = f_x\,dx + f_y\,dy$. For the one-form $\omega = P\,dx + Q\,dy$,

$$d\omega = dP \wedge dx + dQ \wedge dy = (Q_x - P_y)\,dx \wedge dy.$$

The equality follows by writing $dP = P_x\,dx + P_y\,dy$, and using $dx \wedge dx = 0$.

If $f \in C^2$, then

$$d(df) = (f_{yx} - f_{xy})\,dx \wedge dy = 0.$$

This is the local manifestation of $d^2 = 0$. In higher dimensions the corresponding identities include curl of a gradient being zero and divergence of a curl being zero, with the appropriate form identification.

§1.4 Green's theorem in a rectangle

For $R = [a,b] \times [c,d]$ with counterclockwise boundary and a C^1 form $P\,dx + Q\,dy$, calculate each edge:

$$\begin{aligned}\int_{\partial R} P\,dx + Q\,dy = &\int_a^b P(x,c)\,dx + \int_c^d Q(b,y)\,dy \\ &- \int_a^b P(x,d)\,dx - \int_c^d Q(a,y)\,dy.\end{aligned}$$

Apply the fundamental theorem of calculus first in y to the two P-terms and then in x to the Q-terms:

$$\int_{\partial R} \omega = \int_a^b \int_c^d (Q_x - P_y)\,dy\,dx = \int_R d\omega.$$

This calculation reveals the mechanism. If a region is subdivided into rectangles, each interior oriented edge appears twice with opposite signs and cancels. Only the outer boundary survives. For a general smooth domain, parametrization, subdivision, approximation and compatibility on overlaps make that cancellation rigorous.

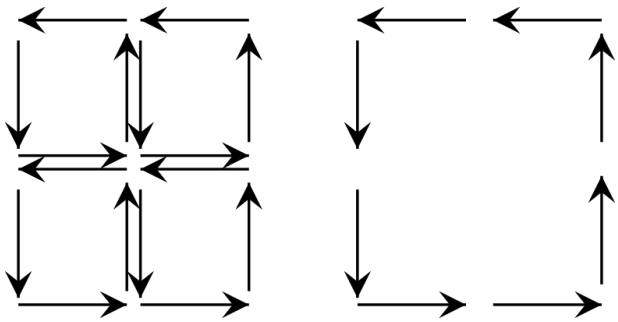


Figure 1.1: Orientation of the patch induces an orientation on the boundary, determining the signs in Stokes' theorem.

§1.5 The exterior derivative commutes with pullback

If $\Phi : U \to M$ is a smooth map and f a smooth function on M, the chain rule says $d(f \circ \Phi) = \Phi^*(df)$. Since both the pullback and d respect the alternating product, the identity extends to arbitrary forms:

$$\boxed{\Phi^*(d\omega) = d(\Phi^*\omega)}.$$

For a two-dimensional parameterization $\Phi(u, v) = (x(u, v), y(u, v))$, one sees the Jacobian directly:

$$\Phi^*(dx \wedge dy) = (x_u y_v - x_v y_u)\, du \wedge dv.$$

The determinant's sign is exactly what records whether a parametrization preserves or reverses orientation. This is the mechanism behind change of variables, rather than an extra correction added to a formula after the fact.

§1.6 Stokes' theorem and induced orientation

Theorem 1.6.1 (Generalized Stokes)

Let M be a compact oriented smooth n-manifold with smooth boundary, and let ω be a smooth $(n-1)$-form defined near M. With the induced boundary orientation,

$$\int_M d\omega = \int_{\partial M} \omega.$$

At a boundary point, the outward normal is placed first: a tangent basis $v_1, \ldots, v_{n-1}$ is positively oriented on ∂M when (outward normal,$v_1, \ldots, v_{n-1}$) is positively oriented on M. In the oriented plane this makes the boundary run counterclockwise. The proof reduces the manifold locally to parameterized pieces in Euclidean space; interior faces cancel as in the rectangle, while a partition of unity controls the local-to-global assembly. We have demonstrated the rectangular computation above, not a complete proof of that manifold theorem.

§1.7 The divergence theorem as a three-form computation

For a vector field $F = (P, Q, R)$ on $\mathbb{R}^3$, its flux form is

$$\eta = P\, dy \wedge dz + Q\, dz \wedge dx + R\, dx \wedge dy.$$

Exterior differentiation yields

$$d\eta = (P_x + Q_y + R_z)\, dx \wedge dy \wedge dz.$$

Thus general Stokes specializes to the divergence theorem. Take $F(x, y, z) = (x, y, z)$ and M the unit ball. The divergence equals 3, so the volume side is

$$\int_M d\eta = 3\frac{4\pi}{3} = 4\pi.$$

On the unit sphere the outward unit normal equals F; the flux density is $F \cdot n = 1$, and the boundary integral is its area, again 4π. In a single example, differential forms and the familiar vector calculus formula give the same computable answer.

§1.8 A closed form can remember a hole

Not every closed form is globally a derivative. On $\mathbb{R}^2 \setminus \{0\}$, consider

$$\alpha = \frac{-y\, dx + x\, dy}{x^2 + y^2}.$$

Differentiation shows $d\alpha = 0$. On the unit circle, however,

$$\gamma(t) = (\cos t, \sin t), \qquad \gamma^*\alpha = dt, \qquad \int_\gamma \alpha = 2\pi.$$

If $\alpha = df$ globally for a single-valued smooth function, the fundamental theorem of calculus would force its integral around every closed curve to be zero. Therefore α is closed but not exact.

Locally $\alpha = d\theta$, where θ is an angular coordinate. One can choose an angle on a small simply connected patch, but no smooth single-valued angular function goes all the way around the missing origin. This is the elementary obstruction measured by the first de Rham cohomology group: the distinction between closed one-forms and exact one-forms records genuine topology.

The generalized Stokes formula is now more than a slogan. A form can be pulled back along a parametrization, its exterior derivative is compatible with that pullback, and induced boundary orientations ensure that the interior contributions cancel. The failure of closed forms to be exact explains why global topology still matters even when the local derivative vanishes.